

Reproducible Data Retrieval: Connecting to the Cloud – Answer Key

In the last few modules, we have cleaned data that were pulled from various sources and you were directed to calculate specific metrics. In this module, you will learn how to pull data for yourself and use that data to calculate metrics that you hypothesize may be important.

Objectives:

1. Navigate the US Census API to locate variables of interest from the American Community Survey.
2. Use the `tidycensus` package to pull data from the American Community Survey and calculate a descriptive metric of interest.
3. Use the NASS Quick Stats page to locate variables of interest from the Census of Agriculture.
4. Use the `rnassqs` package to pull data from the Census of Agriculture and calculate a descriptive metric of interest.

Getting Started

(1) Open RStudio and your project file. (2) Create a new Quarto file.

American Community Survey

In the last module, we used the `tidycensus` package to pull data on the number of retirees based on the code assigned to each piece of information. If you want to pull your own data, you will first need to find the correct codes.

All the codes for the American Community Survey can be found online at <https://api.census.gov/data/2017/acs/acs5/groups.html>. Note that you can substitute 2017 in this URL with any year from 2009-2023. Since we have been working with 2017 data from the Census of Agriculture, we use 2017 here for consistency.

When you go to this webpage, you will see a table with a table code, a table description, and a link to a variable list. First, choose a description that you'd like to take a closer look at. For this example, I'll look at a table with information on citizenship. The description is NATIVITY AND CITIZENSHIP IN THE UNITED STATES and the table code is B05001.

If I want the whole table, I can use `tidycensus` to pull in the whole table.

(3) Import `tidycensus` library.

```
library(tidycensus)
```

(4) Import the Census API key you received in Module 5 with `census_api_key("YOUR API KEY GOES HERE")`. (5) Hide your API key in the report with `{r, include=FALSE}`.

(6) Pull in the nativity and citizenship table with the code below:

```
citizens <- get_acs(geography = "county",
                    state = "ID",
                    county = "Bear Lake", # your county here
                    year = 2017,
                    survey = "acs5",
                    table = "B05001")
```

(7) Print the `citizens` data.

```
print(citizens)
```

A tibble: 6 x 5

```

  GEOID NAME          variable estimate moe
  <chr> <chr>          <chr>      <dbl> <dbl>
1 16007 Bear Lake County, Idaho B05001_001      5942    NA
2 16007 Bear Lake County, Idaho B05001_002      5735     67
3 16007 Bear Lake County, Idaho B05001_003         0     17
4 16007 Bear Lake County, Idaho B05001_004        45     34
5 16007 Bear Lake County, Idaho B05001_005       126     43
6 16007 Bear Lake County, Idaho B05001_006        36     25

```

(8) You should see 5 columns: GEOID, NAME, variable, estimate, moe. GEOID stands for geographic identifier, a unique code that applies to your county. NAME is the name of the GEO ID – the name of your county, in this case. variable is the code indicating the information contained in each row. estimate is the number of people in each category. moe stands for margin of error and estimates how accurate the estimate is. We will not use moe in this analysis.

Now that you have a table, you need to know what those variables are! (9) Click the “selected variables” link for the B05001 table on the American Community Survey API website to see the variable codes.

B05001	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	selected variables
--------	--	--------------------

Census Data API: Variables in /data/2017/acs/acs5/groups/B05001

Name	Label	Concept	Required	Attributes	Limit	Predicate Type	Group
B05001_001E	Estimate!!Total	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	int	B05001
B05001_001EA	Annotation of Estimate!!Total	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	string	B05001
B05001_001M	Margin of Error!!Total	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	int	B05001
B05001_001MA	Annotation of Margin of Error!!Total	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	string	B05001
B05001_002E	Estimate!!Total!!U.S. citizen, born in the United States	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	int	B05001
B05001_002EA	Annotation of Estimate!!Total!!U.S. citizen, born in the United States	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	string	B05001
B05001_002M	Margin of Error!!Total!!U.S. citizen, born in the United States	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	int	B05001
B05001_002MA	Annotation of Margin of Error!!Total!!U.S. citizen, born in the United States	NATIVITY AND CITIZENSHIP STATUS IN THE UNITED STATES	predicate-only		0	string	B05001

Each code appears in this table four times. For example, variable B05001_001 appears as B05001_001E, B05001_001EA, B05001_001M, and B05001_001MA. tidycensus cleaned up all this detail for us, so we can focus on the variables that end in “E” for estimate. This table shows that variable B05001_001 corresponds to the total population. Variable B05001_002 is the number of U.S. citizens born in the United States in your county.

(10) Using the codes listed online as a reference, calculate the percentage of people who are not a U.S. citizen in your county.

```
not_citizen <- subset(citizens, variable == "B05001_006")$estimate
tot_pop <- subset(citizens, variable == "B05001_001")$estimate

not_citizen / tot_pop * 100
```

```
[1] 0.6058566
```

In the last module, we did not pull a whole table into R. Some tables contain much more information than we need. If you just want a few variables, you can click the “selected variables” link next to a table of interest, and then import the only the variable codes you want.

(11) Modify the code below to import two or more variables for your county from any ACS table. Do not add “E”, “EA”, “M”, or “MA” at the end of the variable codes in this function.

```
pulling_variables <- get_acs(geography = "_____",
                             state = "__",
                             county = "_____",
                             year = 2017,
                             survey = "acs5",
                             variables = c("B????_???", "_____"))
# add more variables if you'd like to
```

Optional: (12) Pull data from the American Community Survey and calculate any metrics that you think might be changing in the farming community of your county. You can pull from different tables and different years, and use any coding methods you’ve learned. This is your time to experiment!

If you’d like more information about how you can use `tidycensus`, you can visit their tutorial site at <https://walker-data.com/tidycensus/articles/basic-usage.html>.

Census of Agriculture

There are a few packages similar to `tidycensus` built to pull data from the Census of Agriculture. Like `tidycensus`, you will need to request an API key and find the different data options online.

(13) Install the `rnassqs` package in the Console with `install.packages("rnassqs")`.

(14) Read in the `rnassqs` library.

```
library(rnassqs)
```

(15) Go to <https://quickstats.nass.usda.gov/api/> and click “Request an API key” on the left side of the page to get an API key from the USDA.

(16) Import your API key with the code below. Hide your key with `include=FALSE`.

```
nassqs_auth(key = "YOUR API KEY HERE")
```

The general format of an `nassqs` query is to specify a list of parameters that should apply to the data you want to pull, and then feed that list into the `nassqs` function. For example, to find the number of farms with conservation easements, I’ll first specify a list of parameters that will get me the information I want. (17) Create this list of parameters for your county. I have added a new line for readability for the `short_desc` element, but you will need to replace it with a single space.

```
easements_params <- list(state_alpha = "ID",
                          agg_level_desc = "COUNTY",
                          county_name = "BEAR LAKE",
                          commodity_desc = "GOVT PROGRAMS",
                          short_desc = "GOVT PROGRAMS, FEDERAL,
                                      CONSERVATION & WETLANDS - NUMBER OF OPERATIONS")
```

This will tell the `nassqs` code to retrieve all data from the Census of Agriculture that matches these values.

(18) Pull the data from the Census of Agriculture by using the list of parameters we created as the argument for the function `nassqs`, which pulls data from the Census of Agriculture servers.

```
farms_with_conservation <- nassqs(easements_params,
                                  as_numeric = TRUE,
                                  progress_bar = FALSE)
```

(19) Print your data. There should be 5 rows, with information about the number of farms with conservation easements in 1997, 2002, 2007, 2012, and 2017. Even though the column names are different, the `Value` column still holds the value of interest.

```
print(farms_with_conservation)
```

	source_desc	sector_desc	group_desc	commodity_desc
1	CENSUS	ECONOMICS	FARMS & LAND & ASSETS	GOVT PROGRAMS
2	CENSUS	ECONOMICS	FARMS & LAND & ASSETS	GOVT PROGRAMS
3	CENSUS	ECONOMICS	FARMS & LAND & ASSETS	GOVT PROGRAMS

4	CENSUS	ECONOMICS FARMS & LAND & ASSETS	GOVT PROGRAMS
5	CENSUS	ECONOMICS FARMS & LAND & ASSETS	GOVT PROGRAMS
6	CENSUS	ECONOMICS FARMS & LAND & ASSETS	GOVT PROGRAMS

	class_desc	prodn_practice_desc
1	FEDERAL, CONSERVATION & WETLANDS	ALL PRODUCTION PRACTICES
2	FEDERAL, CONSERVATION & WETLANDS	ALL PRODUCTION PRACTICES
3	FEDERAL, CONSERVATION & WETLANDS	ALL PRODUCTION PRACTICES
4	FEDERAL, CONSERVATION & WETLANDS	ALL PRODUCTION PRACTICES
5	FEDERAL, CONSERVATION & WETLANDS	ALL PRODUCTION PRACTICES
6	FEDERAL, CONSERVATION & WETLANDS	ALL PRODUCTION PRACTICES

	util_practice_desc	statisticcat_desc	unit_desc
1	ALL UTILIZATION PRACTICES		AREA OPERATIONS
2	ALL UTILIZATION PRACTICES		AREA OPERATIONS
3	ALL UTILIZATION PRACTICES		AREA OPERATIONS
4	ALL UTILIZATION PRACTICES		AREA OPERATIONS
5	ALL UTILIZATION PRACTICES		AREA OPERATIONS
6	ALL UTILIZATION PRACTICES		AREA OPERATIONS

	short_desc
1	GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - NUMBER OF OPERATIONS
2	GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - NUMBER OF OPERATIONS
3	GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - NUMBER OF OPERATIONS
4	GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - NUMBER OF OPERATIONS
5	GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - NUMBER OF OPERATIONS
6	GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - NUMBER OF OPERATIONS

	domain_desc	domaincat_desc	agg_level_desc	state_ansi	state_fips_code
1	TOTAL	NOT SPECIFIED	COUNTY	16	16
2	TOTAL	NOT SPECIFIED	COUNTY	16	16
3	TOTAL	NOT SPECIFIED	COUNTY	16	16
4	TOTAL	NOT SPECIFIED	COUNTY	16	16
5	TOTAL	NOT SPECIFIED	COUNTY	16	16
6	TOTAL	NOT SPECIFIED	COUNTY	16	16

	state_alpha	state_name	asd_code	asd_desc	county_ansi	county_code	county_name
1	ID	IDAHO	90	EAST	007	007	BEAR LAKE
2	ID	IDAHO	90	EAST	007	007	BEAR LAKE
3	ID	IDAHO	90	EAST	007	007	BEAR LAKE
4	ID	IDAHO	90	EAST	007	007	BEAR LAKE
5	ID	IDAHO	90	EAST	007	007	BEAR LAKE
6	ID	IDAHO	90	EAST	007	007	BEAR LAKE

	region_desc	zip_5	watershed_code	watershed_desc	congr_district_code
1			00000000		
2			00000000		
3			00000000		
4			00000000		

```

5          00000000
6          00000000
country_code country_name location_desc year freq_desc begin_code
1      9000 UNITED STATES IDAHO, EAST, BEAR LAKE 2022 ANNUAL 00
2      9000 UNITED STATES IDAHO, EAST, BEAR LAKE 2017 ANNUAL 00
3      9000 UNITED STATES IDAHO, EAST, BEAR LAKE 2012 ANNUAL 00
4      9000 UNITED STATES IDAHO, EAST, BEAR LAKE 2007 ANNUAL 00
5      9000 UNITED STATES IDAHO, EAST, BEAR LAKE 2002 ANNUAL 00
6      9000 UNITED STATES IDAHO, EAST, BEAR LAKE 1997 ANNUAL 00
end_code reference_period_desc week_ending load_time Value
1      00 YEAR 2024-07-02 12:00:00.000 37
2      00 YEAR 2018-02-01 00:00:00.000 50
3      00 YEAR 2012-12-31 00:00:00.000 81
4      00 YEAR 2012-01-01 00:00:00.000 130
5      00 YEAR 2012-01-01 00:00:00.000 98
6      00 YEAR 2012-01-01 00:00:00.000 99
CV (%)
1 24.6
2 16.1
3 8.2
4
5
6

```

(20) Identify the number of farms with conservation easements in 2017 and (21) calculate the 20-year change in farms with easements from 1997 to 2017.

```

farms_w_eas_2017 <- subset(farms_with_conservation, year == 2017)$Value
farms_w_eas_1997 <- subset(farms_with_conservation, year == 1997)$Value

print(farms_w_eas_2017)

```

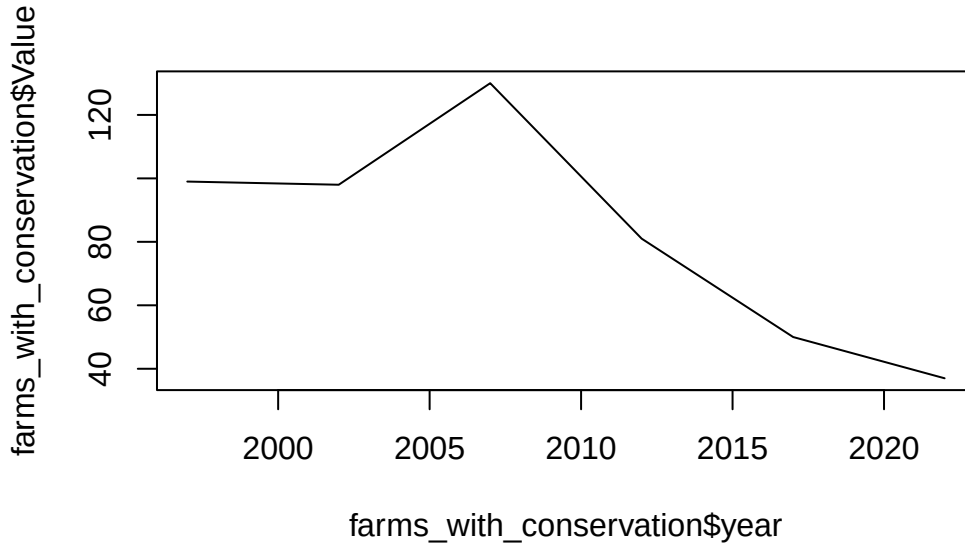
```
[1] 50
```

```
farms_w_eas_2017 - farms_w_eas_1997
```

```
[1] -49
```

Optional: (22) Use `plot` to make a graph of the change in farms with easements over the 20 year period of 1997-2017. Are there any notable trends?

```
plot(x = farms_with_conservation$year,  
     y = farms_with_conservation$Value,  
     type = "l")
```



Finding your own parameters in the Census of Agriculture

You can explore all available parameters at <https://quickstats.nass.usda.gov>. Here, you can narrow down data selection to categories of information, locations, and years. After narrowing down your data selection, you can download a spreadsheet directly from the website. However, it is more reproducible to show the process you used to access data. This will help us track down the data for other counties if you find something interesting!

In Quick Stats, explore the different data categories. **(23)** First, select the “County” option in the “Geographic Level” box, then select “Idaho” in the “State” box and your county in the “County” box. This will make sure that any data options you select will be available for your county.

Select Location (one or more)

Geographic Level:	State:	Ag District:	County:
AGRICULTURAL DISTRICT	HAWAII	EAST	ADA
AMERICAN INDIAN RESERVATION	IDAHO		ADAMS
COUNTY	ILLINOIS		BANNOCK
NATIONAL	INDIANA		BEAR LAKE
PUERTO RICO & OUTLYING AREAS	IOWA		BENEWAH
REGION : MULTI-STATE	KANSAS		BINGHAM
REGION : SUB-STATE	KENTUCKY		BLAINE
STATE	LOUISIANA		BOISE
WATERSHED	MAINE		BONNER

(24) Explore the different kinds of information you can access. Start by selecting an interesting “Sector,” then continuing exploring and narrowing down options until you’ve identified one “Data Item” you’d like to analyze. Below is an example of the exploration I did to find the easements information:

 United States Department of Agriculture
National Agricultural Statistics Service

Quick Stats

Select Commodity (one or more) 

Program:	Sector:	Group:	Commodity:	Category:
CENSUS	ANIMALS & PRODUCTS	FARMS & LAND & ASSETS	FARM OPERATIONS	AREA
	CROPS		FEED	
	DEMOGRAPHICS		FERTILIZER TOTALS	
	ECONOMICS		FUELS	
	ENVIRONMENTAL		GOVT PROGRAMS	
			INCOME, FARM-RELATED	
			INCOME, NET CASH FARM	
			INTEREST	
			LABOR	

Data Item:	Domain:
GOVT PROGRAMS, FEDERAL - OPERATIONS WITH RECEIPTS	TOTAL
GOVT PROGRAMS, FEDERAL - RECEIPTS, MEASURED IN \$	
GOVT PROGRAMS, FEDERAL - RECEIPTS, MEASURED IN \$ / OPERATION	
GOVT PROGRAMS, FEDERAL, (EXCL CONSERVATION & WETLANDS) - OPERATIONS WITH RECEIPTS	
GOVT PROGRAMS, FEDERAL, (EXCL CONSERVATION & WETLANDS) - RECEIPTS, MEASURED IN \$	
GOVT PROGRAMS, FEDERAL, (EXCL CONSERVATION & WETLANDS) - RECEIPTS, MEASURED IN \$ / OPERATION	
GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - ACRES	
GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - NUMBER OF OPERATIONS	
GOVT PROGRAMS, FEDERAL, CONSERVATION & WETLANDS - OPERATIONS WITH RECEIPTS	

To pull the data item you want, you will need to use the language of `rnassqs`. You can see a full description of every option by running `?nassqs` in the Console. (25) Use the values you selected in Quick Stats to make a list of parameters.

```
my_parameters <- list(state_alpha = "ID",
                      agg_level_desc = "COUNTY",
                      county_name = "YOUR COUNTY",
                      year = YYYY,
                      sector_desc = "", # Sector
                      group_desc = "", # Group
                      commodity_desc = "", # Commodity
                      statisticcat_desc = "", # Category
                      short_desc = "", # Data Item
                      domain_desc = "") # Domain
```

You don't need to use all of these options. For example, in step (17), I didn't use the `year` argument, so I got all the available years.

Optional: (26) Use the parameter list you created in step (25) in the function `rnassqs` to pull the data that matches your parameters into R. Calculate any metrics that you think might be changing in the farming community of your county. You can pull from different parameters and different years, and use any coding methods you've learned. This is your time to experiment!

If you'd like more information about how you can use `rnassqs`, you can visit their tutorial site at <https://cran.r-project.org/web/packages/rnassqs/vignettes/rnassqs.html>.

Finishing up

(27) Summarize the information you learned about your county as a conclusion to this report.

(28) Go back through your report and add short explanations for what each code chunk does in your own words if you haven't done so already. **(29)** Render your report to a PDF and email it to [INSERT EMAIL HERE].

Statement of original and referenced work:

The entirety of this module is original work authored by Carolyn Koehn.

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